

PETER NGUYEN

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EDUCATION

California Polytechnic University, Pomona

Bachelor of Science in Computer Science, Minor in Data Science

Pomona, CA

May 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS, C++, Dart

Frameworks & Libraries: React.js, Next.js, Flutter, TensorFlow, Keras, pandas, NumPy, Matplotlib

Tools & Platforms: Git, GitHub, AWS, Firebase, Supabase, PostgreSQL, MongoDB, Vercel, Postman

Practices: REST APIs, Agile, OOP, Data Structures & Algorithms

EXPERIENCE

HAPII Lab

Data Analyst Intern

Pomona, CA

September 2023 – June 2024

- Built a full-stack survey application comparing three recommender system algorithms using a **JavaScript** frontend and **Firestore** backend to support machine learning research on user preference data
- Collected cross-cultural dataset from 1,000+ international participants via **Amazon Mechanical Turk**, implementing validation and quality assurance protocols to ensure data integrity
- Improved algorithm recommendation accuracy by 15% by conducting statistical analysis in **Python** with **pandas** and **Matplotlib** to identify and act on geographical preference patterns

PROJECTS

Hextech Insight | [GitHub](#) | *React, JavaScript, Supabase, PostgreSQL, Riot Games API, CSS*

- Built a League of Legends analytics platform querying 20+ match data points across 12 global regions by integrating the **Riot Games API** with real-time data sync and a **PostgreSQL** database backend
- Delivered a responsive web app with complex API data processing and visualization using **React** component architecture, **JavaScript** state management, and robust error handling
- Implemented secure user authentication with OAuth login and session management via **Supabase** and **Google** provider integration

Conscious Cart | [GitHub](#) | *Gemini, Flutter, Dart, Python, Firebase*

- Built a sustainable shopping recommendation system processing 500+ nutritional data points from **USDA FoodData Central** using **Gemini API** for AI-powered analysis and **Flutter** for cross-platform deployment
- Developed custom allergen detection logic identifying major allergen categories across food products with optimized filtering and data processing workflows
- Submitted to **Google AI Hackathon** combining machine learning recommendations, **Firestore** backend, and an intuitive mobile UI

Animal & Insect Image Classifier | [GitHub](#) | *Python, TensorFlow, Keras, Tkinter*

- Achieved 85% classification accuracy across 10 species by applying **MobileNetV2** transfer learning with custom preprocessing pipelines on a 10,000+ image dataset
- Improved model performance through data augmentation and hyperparameter tuning, reducing overfitting on limited training samples
- Shipped a **Tkinter** desktop app with drag-and-drop image input and real-time prediction display for end-user accessibility

Chess Neural Network | [GitHub](#) | *Java, Java Swing, Machine Learning*

- Trained a chess AI on 1,500+ professional games with position evaluation algorithms to produce strategic, competitive gameplay
- Implemented a complete chess engine with full rule enforcement, move validation, and a responsive **Java Swing** GUI using object-oriented design principles